Ryan Hausmann

ryanhausmann3@gmail.com | (848) 234-4778 | Keyport, NJ | github.com/ryanhaus

EDUCATION

B.S. in Electrical and Computer Engineering September 2023 - May 2027 (expected)
Rowan University — Glassboro, NJ Cumulative GPA: 4.0/4.0

Minor in Computer Science, President's Scholar of Excellence, John H. Martinson Honors College, Baja SAE Team, Engineering Learning Community Mentor, Recipient of 2x Rowan Scholars Program scholarships.

WORK EXPERIENCE

Optical Hardware Development Intern, Nokia, *Murray Hill, NJ* May - August 2025

- Created and tested simulation models of a TCA9539 I^2C I/O expander and an AT24C02 I^2C EEPROM in SystemVerilog, as well as a controller module which generated I^2C signals to interface with the model. Verified compatibility with existing testbench infrastructure.
- Performed board bring-up for multiple fan controller boards and wrote software in Python to validate all functionalities of the board. Performed rework where necessary to get the boards to a working state.
- Identified and fixed a bug in the open-source LiteX project which prevented booting over TFTP in the BIOS.
- Created and tested a Universal Program and Debug Interface (UPDI) programmer in SystemVerilog, which was able to flash a compiled program to a connected microcontroller on an Arty A7-100T board.

Optical Hardware Development Intern, Nokia, *Murray Hill, NJ* June - August 2024

- Used LiteX to generate a Linux-capable SoC on an Artix-7 FPGA, wrote drivers to interface with logic.
- Designed and built a temperature-controlled AC power cutoff device to prevent thermal runoff during tests.
- Designed a PCB and wrote software for a common redundant power supply (CRPS) data logging device.
- Performed thermal tests on QSFP-DD transceiver modules, analyzed and organized results in Excel.

Software Development Mentee, Commvault, *Tinton Falls, NJ (remote)* January - June 2023

- Collaborated with a mentor to develop file virus scanning software in C++.

PROJECTS

Personal "Home Lab" January 2026 - present

- Set up a custom "home lab" using 4x Raspberry Pis running Linux, used Docker to containerize services.
- Modified a 3D-printed enclosure to hold the Pis, a network switch, power supply, and a hard drive.

Rowan University's Baja SAE Data Acquisition System September 2025 - present

- Designed a custom PIC32-based PCB to connect various sensors around a Baja vehicle via CAN Bus.
- Hand-assembled the PCBs, which has components as small as 0402 and a QFN-48 component.
- Wrote software in C and created an automated system with CMake to compile programs for different targets.

FPGA Triangle Rasterizer & 3D Renderer January 2025

- Wrote a simple triangle rasterizer in SystemVerilog and accompanying simulation software.
- Implemented a 3D renderer using the rasterizer with z-buffers for proper rendering.

Intel 8080 CPU Emulator November 2024

- Wrote an emulator for the 8-bit Intel 8080 microprocessor in the Rust programming language.
- Thoroughly tested each CPU instruction in the emulator with publicly available test programs.

SKILLS

Hardware: PCB Design with KiCad, Altium, Eagle. Experienced with soldering and component rework. Experience using oscilloscopes, logic/protocol analyzers, multimeters, function generators, etc.

FPGA: Design & verification with (System)Verilog using Vivado, Quartus, Verilator, Yosys, and LiteX.

Software: Programming with C, C++, Rust, JavaScript, Java, Python, shell scripts, and MATLAB. Experience with Linux (Fedora, Ubuntu, embedded, etc.), Vim, Git, CMake, and Docker.

Design: CAD with OnShape and Fusion 360. Experience with rapid prototyping with 3D printing.